



OPEN De novo transcriptome assembly and differential gene expression analysis in different developmental stages of *Agriotes sputator* (click beetle)

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Wireworms, the larva of click beetle (*Agriotes* species), are one of the most destructive pests of horticultural crops in North America, responsible for considerable economic losses in Canada. *Agriotes sputator* (*A. sputator*) species is a predominant wireworm pest attacking potato fields in Eastern Canada. However, no information about its genome-wide gene expression profile, specifically for the genes involved with development is available to date. Therefore, we generated the transcriptome profile of *A. sputator* during five developmental stages, including the three larval stages and adult male and female click beetle. Out of 714.7 million raw reads, *de novo* assembly generated 564,561 transcripts. The data were subjected to differential expression analysis using DESeq2, gene ontology, annotation, and pathway analyses. A total of 34,709 differentially expressed genes (DEGs) were significant ($\log_2$ fold change > 2, $\text{padj} < 0.05$) across the developmental stages. Functional analysis of DEGs identified development signaling, metabolism, transport, cellular mechanisms, and drug metabolism (cytochrome p450) pathways. This study provides comprehensive sequence resources and potential gene differences at different developmental stages of *A. sputator*. These findings will represent a major step towards developing sustainable methods to control this widely distributed pest in agricultural fields.

Keywords Click beetle, Wireworm larvae, RNA-Seq, Transcriptome analysis, Different developmental stages and differentially expressed genes

Wireworms (Coleoptera: Elateridae) are the larval stage of click beetles (*Agriotes* species) originating from Europe and negatively impacting potato production in Canada¹. Wireworms consume perennial parts of plants, contributing to deformed or damaged crops and leading to an increasing amount of unmarketable produce². The biology and ecology of some *Agriotes* species have been thoroughly studied^{3–7}. Among *Agriotes* species, three destructive European species, *Agriotes obscurus*, *A. lineatus*, and *A. sputator*, became dominating in Atlantic Canada over the past 10 years, particularly affecting potato crops¹. The wireworm life cycle using instar has not been clarified⁸. It is understood that wireworms have high sustainability and a multi-annual lifecycle by living up to six years in the soil before emerging in the spring as mature click beetles⁸. Among the most popular management practices and preventive measures to control pest infestations, chemical pesticides have been the most common approach. However, the overuse of pesticides adversely affects food and the environment. Therefore, there is a high demand for an alternate and safe pest control strategy for wireworms.

In recent years, RNA interference (RNAi) method, also known as double-stranded RNA (dsRNA)-mediated gene silencing, has shown promising results in techniques for controlling Coleoptera pests⁹ because interference with critical insect genes using RNAi can lead to the death of pests. Proof of principle for the application of RNAi in insect crop pest control comes from early studies conducted on the western corn rootworm (*Diabrotica virgifera*)¹⁰, and cotton bollworm (*Helicoverpa armigera*)¹¹. Finding appropriate target genes is essential to developing RNAi-mediated pest control strategies. Non-target species should not be affected by those RNAi target genes. Unfortunately, the genetic resources for this insect are very limited, additional resources are needed in order to effectively screen target genes.

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High-throughput sequencing technology, especially RNA sequencing (RNA-Seq), has obtained transcriptome data from non-model species, providing valuable functional, comparative, and evolutionary genomics information, even without genome sequences^{12–14}. RNA-Seq information can effectively help to understand cellular responses, gene function and the molecular mechanisms behind various biological processes, such as insect feeding mechanism¹⁴, defense¹⁵, and development process¹⁶. Several studies using transcriptomic-based approaches have focused on identifying genes involved in developmental processes, with expression profiles among different life stages reported in several insect species^{16–18}. Over past years, insect transcriptomes have been reported to be useful genetic resources for high-throughput screening of RNAi target genes in other insect pests¹⁹.

Even though wireworm causes significant damage to economic crops in Canada, there is no high-quality transcriptome and annotations for wireworm of click beetle. The lack of sequence-based information has hindered progress in understanding the wireworm's biology and proposing alternative control methods to pesticide use. Furthermore, the RNAi approach could not be applied in wireworms because of a lack of genomic information.

To accomplish our goals, we conducted the *de novo* transcriptome study of *A. sputator* using Illumina paired-end technology. The sequencing included the five developmental stages of the insect's life cycle, including three larval stages and adult male and female click beetle. To our knowledge, this is the first report on the transcriptomic profiling of *A. sputator*. Genes involved in metabolic processes, general development and reproduction were identified and functionally annotated. Many differentially expressed genes (DEGs), several key Gene Ontology (GO) terms, pathways were also obtained during various periods of insect growth. Using the sequencing technology, this study provides a foundation for future research on developing RNAi of target genes as a strategy for the management of this pest.

Results

Generation and assembly of wireworm and click beetle transcriptomes

For the transcriptome assembly and analysis, a total of 20 RNA samples were used. Out of 20 RNA samples, one sample of early-stage larva (LE4) was not included because the sample was degraded. So, 19 RNA-Seq libraries were generated using the Illumina NEBNext Ultra II RNA library preparation kit, to produce 100 base paired-end reads. Table 1 summarizes sample statistics for all RNA-Seq reads.

In total, 714.7 million pairs of 100 bp paired-end raw reads were generated from five developmental stages of *A. sputator*, and the GC contents were more than 43% across all stages (Table 1). Within each stage, there were more than 26 million raw reads (Table 1). Prior to assembly reads into transcripts, raw reads were quality assessed

Sample	Raw reads	Clean reads	GC content (%)	Mapped to assembled transcript (%)
LE1	46,939,427	45,508,646	44	99.16
LE2	34,293,930	33,758,607	44	99.17
LE3	60,524,968	59,723,195	44	99.34
LM1	31,827,183	31,166,221	44	98.87
LM2	27,665,177	27,009,884	45	99.00
LM3	34,671,712	34,102,967	44	99.12
LM4	41,244,155	40,685,194	44	99.22
LL1	30,883,726	30,363,155	44	98.92
LL2	32,494,661	31,928,930	44	99.10
LL3	28,405,603	27,919,091	44	98.79
LL4	37,305,935	36,731,249	44	99.03
AF1	37,680,932	37,147,015	43	99.28
AF2	32,603,106	32,171,717	43	99.21
AF3	40,097,635	39,591,670	43	99.32
AF4	26,590,991	26,185,121	43	98.76
AM1	30,961,215	30,541,375	44	98.99
AM2	45,739,503	45,057,881	44	99.15
AM3	65,202,450	64,050,073	44	99.22
AM4	29,521,398	29,117,347	45	98.96
Total	714, 653,707	702,759,338		

Table 1. Summary of quality of RNA-Seq original sequencing data. The raw read column lists each read in the FASTQ format file and counts the number of sequences in each file. The clean read column lists the filtered sequencing data of raw reads, and the subsequent bioinformatic analysis is based on clean reads. GC content column, i.e., the percentage of G + C, and mapped to assembled transcripts column generated transcripts from each sample mapped back to the clean reads using Bowtie2 software. Sample LE4 was not included because of the low RNA integrity number (RIN). LE, LM, LL, AF, and AM refer to an early stage of larva, mid-stage of larva, late stage of larva, and adults of female and male click beetle, respectively.

using the FastQC tool and this showed a high Phred quality score (average score over all sequences > 35 Phred) but indicated some bias in the first 10 bases in reads. To determine whether this potential bias was due to adaptor sequences that were left from the sequencing process or to low-quality bases, we applied the Trimmomatic software to clean the reads. As a result, Trimmomatic trimmed adaptor sequences from the reads and removed low-quality bases. Therefore, 702.8 million pairs of clean reads (98.33%) were retained and used for the assembly (Table 1). Using the Trinity assembly program, all clean reads from the 19 libraries were combined to generate a single combined reference transcriptome assembly of *A. sputator*. Trinity generated 564,561 transcripts with N50 length of 2,058 bp and an average transcript size of 965 bp (Table 2). Transcript length ranged from 177 to 83,047 bases, with 323,955 transcripts being > 500 bases and 13,225 being > 10 kb (Table 2). The GC content of the final assembly was 38.75% (Table 2), which is close to the GC content of the raw reads (43–45%) (Table 1). To assess the quality of the assembly, the assembled transcriptome were mapped back to the clean reads (Table 1) and demonstrated a high rate of mapping onto the assembly (almost 98%).

Annotation of predicted sequences

Annotation using NCBI-BLASTx against the NR database showed the most hits species against coleopteran species; *Tribolium castaneum* with 57%, second top blast-hit taxa was *Dendroctonus ponderosae* with 12% similar genes with an e-value cut-off of 1E-5. The remaining blast hits showed similarities to other insects in most cases. Out of 564,561 transcripts, a total of 57,994 predicted genes had at least one hit against the NR database, while quite a large proportion of transcripts had no significant match to any of the sequences in the searched database. This should be expected, because there is no sequence information of *A. sputator* or its closely related species in any databases.

Additionally, the mitochondrial cytochrome oxidase subunit 1 (*mtCOI*) gene was retrieved from the RNA-Seq data from all five groups of samples. Using ClustalW (Bioedit), the RNA sequences from the present study were aligned to a reference sequence of the partial *mtCOI* gene (NCBI Accession number: KM445546.1) as well as aligned with the other 11 different *Agriotes* species (Supplementary Fig. 1) to verify the species of *Agriotes*. The result revealed that all the five groups of samples used in this study were *A. sputator* with 98% identity. Additionally, a phylogenetic tree based on the maximum likelihood method revealed one cluster of all five groups of samples with a reference *A. sputator* species (Supplementary Fig. 2).

Identification of DEGs in different developmental stages

Using our reference assembly, we estimated the transcripts abundance using Salmon software to investigate the global transcriptional differences across different developmental stages. The relative abundance of each transcript or gene was reported as transcripts per kilobase million (TPM). Then, the differential expressed gene (DEGs) analysis was performed using the DESeq2 (R package) to analyze the differential expression between comparable samples. Before DEGs analysis, the read counts were filtered out from the assembly based on their TPM metrics using edgeR (R package) and these read counts then were used to estimate the relative abundance of transcripts across all developmental stages. To identify DEGs during the different development stages of *A. sputator*, the female libraries were used as a reference to fully illustrate the two main comparisons in this work, between larval stages (three larva stages based on their sizes; LE: early-stage, LM: mid-stage and LL: late-stage larva) and adult sexes [female (AF) and male (AM) click beetles]. A volcano plot was created to show the overall distribution of down-regulated DEGs (0 to -10 of log₂ fold change) represented on the left panel as well as up-regulated DEGs (0 to 10 of log₂ fold change) on the right panel at different developmental stages (Fig. 1). Four main groups of DEGs (LE vs. AF; LM vs. AF; LL vs. AF, AM vs. AF) were generated (Fig. 1). The DEGs were expressed with fold change values ≥ 2 and adjusted p-value (padj) ≤ 0.05 were considered as significant and represented in red dots in the volcano plots. The principal component analysis (PCA) of gene expression in different developmental stages of *A. sputator* is shown in Fig. 2. The PCA scatter plot showed clusters of each developmental stage and variance of the biological replicates. According to the PCA analysis, three clusters were formed; the AF stage and AM stages formed two different clusters, and all larval stages (LE, LM and LL) formed one separate cluster.

A total of 34,709 transcripts were significantly (padj < 0.05, log₂ fold change ≥ 2) differentially expressed across the five evaluated developmental stages of *A. sputator*, where 7,712 were found in LE stage, 9,219 in LM stage, 9,554 in LL stage, and 8,224 in AM stage. Out of 34,709 transcripts, a total of 23,685 were found to be

Trinity <i>de novo</i> assembly	
Number of transcripts	564,561
Total assembled bases	545,055,635
Average transcript length (bp)	965
N50	2,058
Minimum transcript length	177
Maximum transcript length	83,047
Transcript > 500 bases	323,955
Transcript 500–999 bases	99,966
Transcript 1 kb – 2.9 kbases	100,602
Transcript 2.9 kb – 4.9 kbases	26,813
Transcript 4.9 kb – 10 kbases	13,225
Number of predicted genes	57,994
GC content (%)	38.75

Table 2. Summary statistics of the combined *de novo* transcriptome assembly of *A. sputator*.

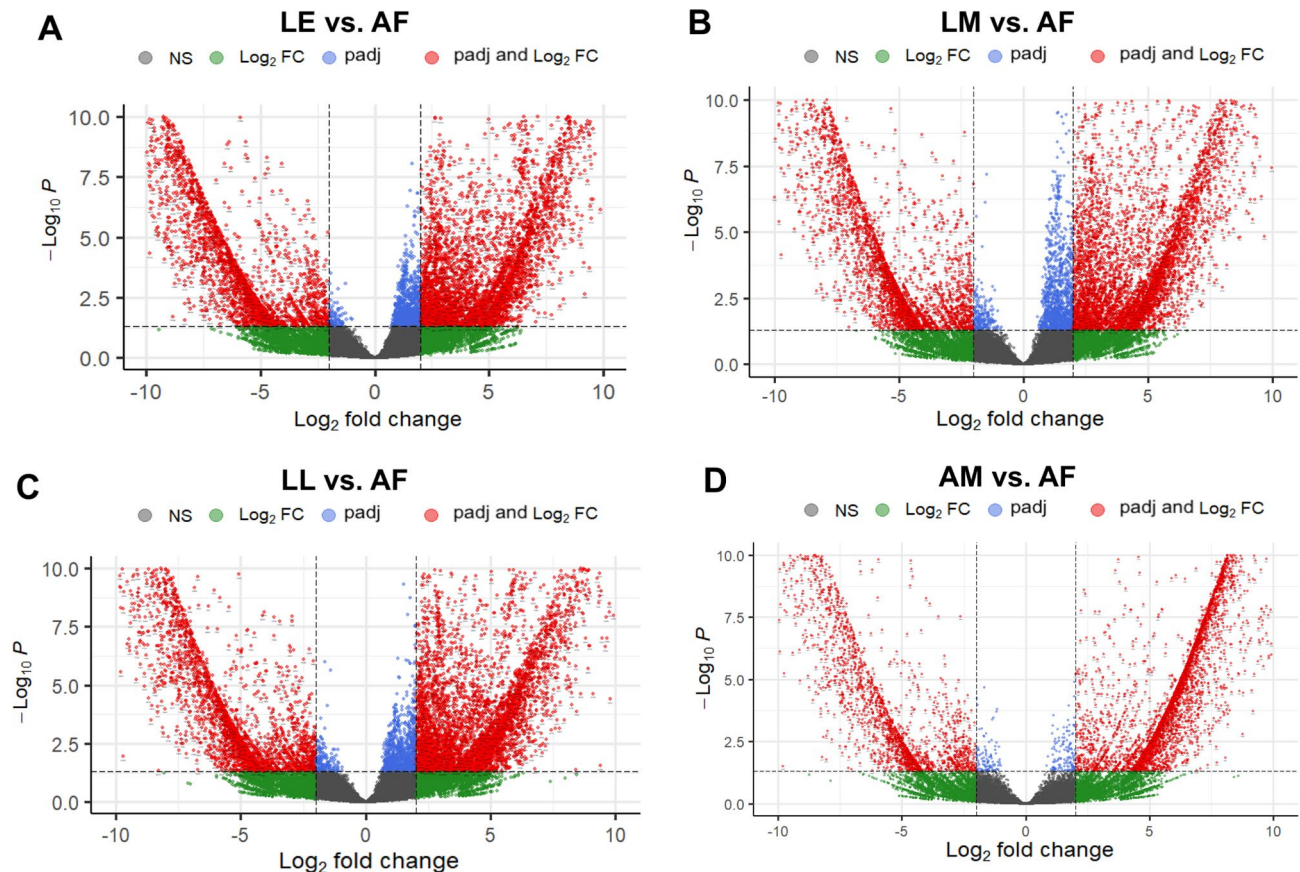


Fig. 1. Volcano plots of DEGs among developmental stages of *A. sputator*. Each dot represents a gene. The x-axis represents the $\log_2$ fold change, and the negative p-value's log (base 10) is on the y-axis. (A) The DEGs between the LE vs. AF stages; (B) the DEGs LM vs. AF stages; (C) the DEGs LL vs. AF stages and, (D) the DEGs AM vs. AF stages. The two vertical and one horizontal dotted grey lines indicate the threshold fold change values ≥ 2 and $\text{padj} \leq 0.05$, respectively. The dots on the left panel (0 to -10 of $\log_2$ fold change) are downregulated genes, while the dots on the right panel (0 to 10 of $\log_2$ fold change) are upregulated genes. Grey dots represent nonsignificant genes (NS) that did not cross the threshold of padj or fold-change; green dots represent the genes with $\log_2$ fold change ≥ 2 ; blue dots represent the genes with significant padj value; and red dots represent significant DEGs ($\text{padj} \leq 0.05$ and $\log_2$ fold change ≥ 2). LE, LM, LL, AF, and AM refer to an early stage of larva, mid-stage of larva, late stage of larva, and adults of female and male click beetle, respectively.

homologs in the NR database while 11,024 were found to be novel genes. Using the adult female (AF) stage as a reference, 4,596, 5,638, 5,950, and 5,801 transcripts were up-regulated, while 3,116, 3,581, 3,604, and 2,423 transcripts were down-regulated in the LE, LM, LL and AM stages, respectively. A full list of significant DEGs among the four different developmental stages against adult female click beetle is summarized in Supplementary Tables 1–4 along with the functional annotation.

Genes and functional pathways involved in the LE stage

Based on 63 GO terms from the DEGs, 26 GO terms were associated with biological process (BP), 18 with cellular component (CC), and 19 with molecular function (MF). The GO terms of each category were sorted according to the number of gene counts, and a bar chart was drawn (Fig. 3A). In the BP group, the top GO terms (from large to small gene counts) belonged to the multicellular organism reproduction, proteolysis, sensory perception of smell, glutathione metabolic process, and lipid catabolic process (Fig. 3A). In the CC group, most of the DEGs are present in the cytoplasm, extracellular space, cytosol and many other compartments (Fig. 3A). In the MF group, the DEGs are involved in heme binding, monooxygenase activity, iron ion binding, oxidoreductase activity and many more (Fig. 3A). The KEGG enrichment analysis showed that 14 pathways were enriched with p-value < 0.05 (Fig. 4A). The top pathways from large to small gene counts in Fig. 4A, are involved in the metabolic pathways, valine, leucine and isoleucine degradation, cysteine and methionine metabolism, cellular mechanism (lysosome), biosynthesis of amino acid, carbon metabolism, oxocarboxylic acid metabolism, glycine, serine and threonine metabolism, motor proteins, fatty acid metabolism and tryptophan metabolism. All the GO terms, KEGG pathways and other annotation information of all significant DEGs found in the LE stage are represented in Supplementary Table 1.

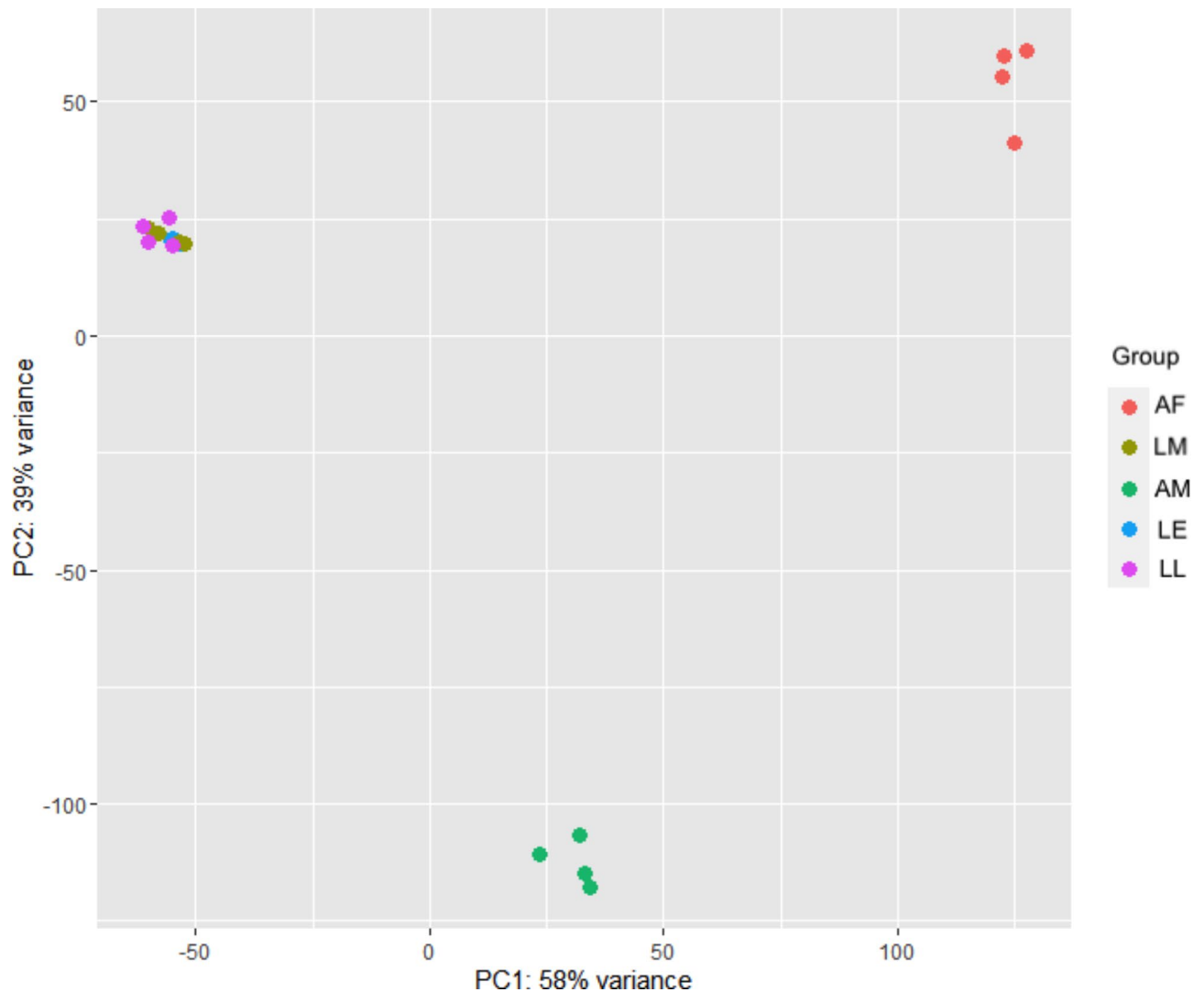


Fig. 2. PCA plot of DEGs in different developmental stages of *A. sputator*. PCA scatter plot was obtained using normalized RNA-Seq data of DEGs and each dot represents a biological replicate of an RNA-Seq sample. It showed clusters of each developmental stage and variance. The percentages on each axis represent the percentages of variation explained by the principal components. LE, LM, LL, AF, and AM refer to an early stage of larva, mid-stage of larva, late stage of larva, and adults of female and male click beetle, respectively.

Genes and functional pathways involved in the LM stage

A total of 60 GO terms were found from the DEGs where 24 GO terms were associated with BP, 17 GO terms with CC and 19 GO terms with MF. The GO terms of each category were sorted according to the number of gene counts. A bar chart was drawn as shown in Fig. 3B. In the BP group, the top GO terms (from large to small gene counts) belonged to the sensory perception of smell, proteolysis, multicellular organism reproduction, glutathione metabolic process, cell division and lipid catabolic process. At the same time, most of the DEGs are present in the cytoplasm, extracellular space, cytosol and many other compartments in the CC group (Fig. 3B). Metal-ion binding, heme binding, iron ion binding, monooxygenase activity, oxidoreductase activity, serine-type endopeptidase activity and many others were significantly associated with the MF group (Fig. 3B). Furthermore, KEGG enrichment analysis showed 19 pathways with corrected p-value < 0.05 which is shown in Fig. 4B. The top significant KEGG pathways based on p-value (< 0.05) from small to large, were primarily involved in metabolic pathways, drug metabolism - other enzymes, drug metabolism - cytochrome P450, metabolism of xenobiotics by cytochrome P450, cysteine and methionine metabolism, lysosome, valine, leucine and isoleucine degradation, biosynthesis of amino acids, beta-Alanine metabolism, carbon metabolism, tryptophan metabolism and fatty acid metabolism (Fig. 4B). All the GO terms, KEGG pathways and other annotation information of all significant DEGs found in the LM stage are represented in Supplementary Table 2.

Genes and functional pathways involved in the LL stage

With GO terms identified from DEGs in the LL stage, 57 GO terms (22 with BP, 16 with CC, and 19 with MF) were significantly enriched. The GO terms of each category were sorted according to the number of gene counts.

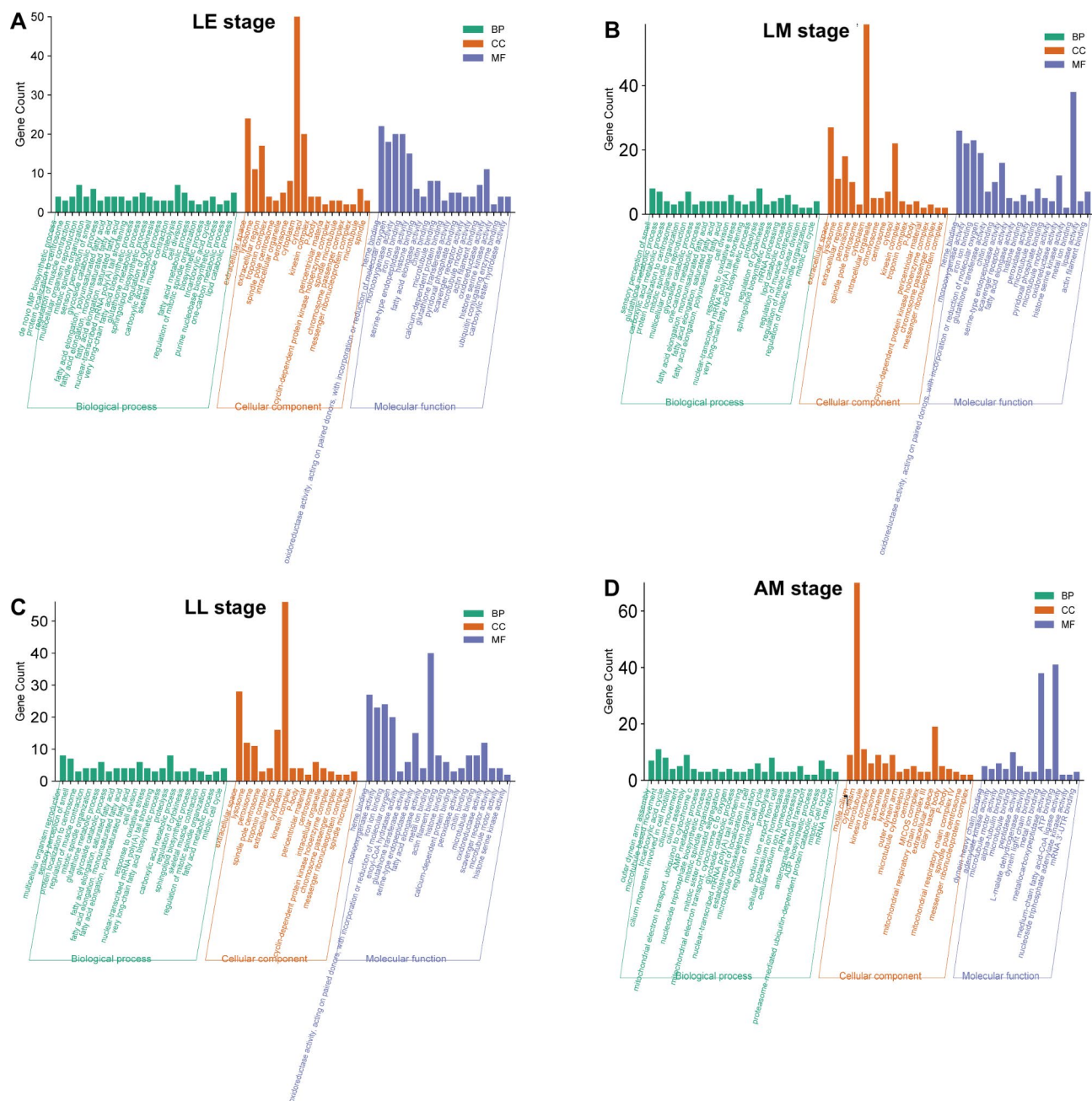


Fig. 3. GO terms for DEGs in each developmental stage. **(A)** GO functional enrichment analysis of DEGs in LE vs. AF. **(B)** GO functional enrichment analysis of DEGs in LM vs. AF. **(C)** GO functional enrichment analysis of DEGs in LL vs. AF. **(D)** GO functional enrichment analysis of DEGs in AM vs. AF beetles. LE, LM, LL, AF, AM, BP, CC and MF refer to an early stage of larva, mid-stage of larva, late stage of larva, adult female, adult male click beetle, biological process, cellular component, and molecular function, respectively. The bar charts were constructed through the bioinformatics platform (SRplot).

The top GO terms (from large to small gene counts) in the BP group are related to the proteolysis, sensory perception of smell, glutathione metabolic process, cell division, regulation of muscle contraction, and fatty acid elongations (saturated, monosaturated, polyunsaturated fatty acid), response to oxidative stress, whereas most of the DEGs in the CC group are involved in cytoplasm, extracellular space and region, lysosome and peroxisome (Fig. 3C). In the MF group, GO terms are related to metal ion binding, heme binding, iron ion binding, monooxygenase activity, serine-type endopeptidase activity and oxidoreductase activity and actin filament binding (Fig. 3C). A KEGG enrichment analysis revealed that the DEGs are classified into 20 pathways with corrected p-value < 0.05 (Fig. 4C). Top enriched pathways, sorted from small to large based on corrected p-value, included metabolic pathways, valine, lysosome, fatty acid metabolism, drug metabolism - other enzymes, glyoxylate and dicarboxylate metabolism, cysteine and methionine metabolism, carbon metabolism, tyrosine

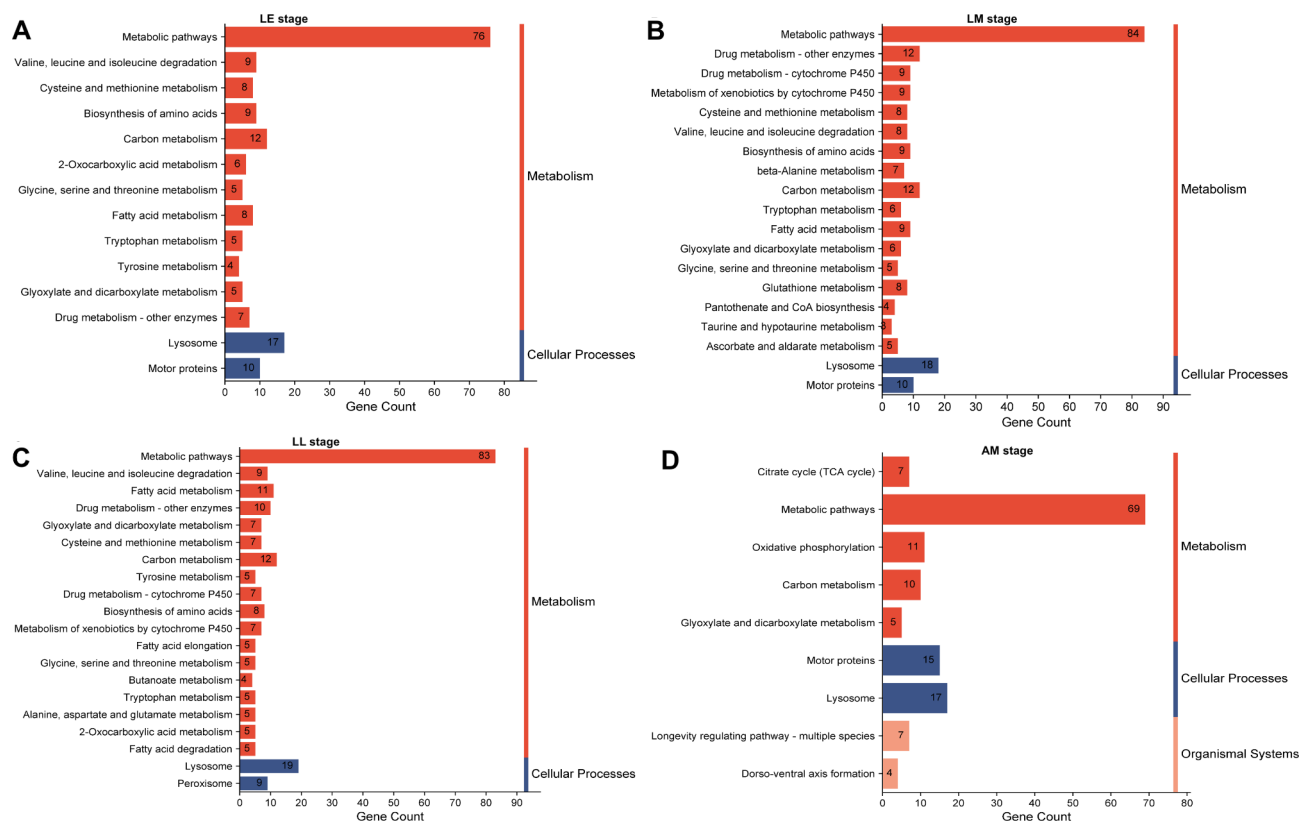


Fig. 4. KEGG enrichment pathway analysis of DEGs based on the comparison among the different developmental stages. **A.** Bar chart of KEGG enrichment results in LE. **B.** Bar chart of KEGG enrichment results in LM. **C.** Bar chart of KEGG enrichment results in LL. **D.** Bar chart of KEGG enrichment results in AM. The vertical axis represents the KEGG pathways, while the horizontal axis represents the number of genes located in the pathway among differential genes to the total number of genes located in the pathway among all annotated genes. LE, LM, LL, AF, and AM refer to early stage of larva, mid-stage of larva, late stage of larva, and adults of female and male click beetle, respectively. The bar charts were constructed through the bioinformatics platform (SRplot).

metabolism, drug metabolism - cytochrome P450, biosynthesis of amino acids, metabolism of xenobiotics by cytochrome P450, fatty acid elongation, glycine, and serine and threonine metabolism (Fig. 4C). All the GO terms, KEGG pathways and other annotation information of all significant DEGs found in the LL stage are represented in Supplementary Table 3.

Genes and functional pathways involved in the AM stage

The DEGs were annotated with 59 GO terms (27 in BP, 18 in CC, and 14 in MF). Several DEGs in BP were significantly involved in the microtubule-based movement, translation process, cilium assembly, proteolysis, tricarboxylic acid cycle (TCA), outer dynein arm assembly, proteasome-mediated ubiquitin-dependent protein catabolic and microtubule cytoskeleton organization, whereas most of the DEGs in the CC group are involved in cytoplasm, extracellular space, microtubule and lysosome (Fig. 3D). In the MF group, ATP binding, metal ion binding, microtubule binding, and microtubule motor activity (Fig. 3D). Meanwhile, KEGG enrichment analysis showed that 28 pathways were enriched with corrected p -value < 0.05 (Fig. 4D). The significantly enriched KEGG pathways were primarily related to the motor proteins, lysosome, TCA cycle, metabolic pathways, oxidative phosphorylation, and longevity regulating pathway, which are the top enriched pathways found in adult male click beetle (Fig. 4D). All the GO terms, KEGG pathways and other annotation information of all significant DEGs found in the AM stage are represented in Supplementary Table 4.

Discussion

Wireworms are the soil-dwelling larvae of click beetles and they are accounted as one of the most severe and widespread agricultural pests worldwide. Larvae attack seeds and developing seedlings, as well as the roots and underground stems of older plants. They can overwinter as either adults or larvae. Throughout the season, wireworms move up and down in the soil depending on temperature. In the winter, they can survive up to two feet deep and as the temperature rises, worms emerge through the ground to the top two to three inches of soil. Once wireworms achieve full maturity during the summer, they pupate in the soil, and within a few days the pupae develop into click beetles. Since the developmental period of wireworms is two to six years, the larvae

can be found in the soil in various sizes and stages throughout the year which makes wireworms a difficult insect pest to deal with.

Popular pest control mechanisms include the use of pesticides; however, pesticides pose a threat to humans and the environment. Therefore, alternate management strategies are demanded to control the wireworm manifestations. Transcriptomics and subsequent RNAi offer promises to identify and silence potential gene targets regulating the growth and development of the wireworms. To date, there is no published data on genome annotation and transcriptomic profiling for wireworm and click beetle (*A. sputator*) except its mitochondrial genome. This shortcoming has prevented research on studying wireworm's life cycle and developing their control strategies for agriculture and food production. Therefore, the study of the genomic structure, transcriptome profiles, or proteome of wireworms to unravel the gene regulations in their life cycle and developmental regulations is needed.

For transcriptomic profiling, cDNA libraries from five different developmental stages of *A. sputator* (LE, LM, LL, AF, and AM) were sequenced. A total of 714.7 million raw data was obtained from Illumina paired-end sequencing technology and then 702.8 million clean reads remained after the filtration. Clean reads were assembled, and 564,561 transcripts were obtained. The quality assessment of the generated assembly, the clean reads from each sample were successfully mapped back to the assembly, indicating a good quality assembly that could be used for further analysis. The GC content of the final assembly was 38.75%, which is close to the GC content of the raw reads (43–45%). *De novo* transcriptome assemblies generally generate significantly more transcripts than a typical annotation^{20,21}, especially at high coverage. The transcriptome dataset that we generated is the most comprehensive to date.

We identified 57,994 predicted genes that matched against the searched database. The unannotated transcripts represent novel genes whose function has not yet been identified. The predicted genes from the *A. sputator* transcriptome showed the most BLAST hits with *T. castaneum* (red flour beetle) probably because (i) their similar biological genomic conservation as both species are classified under the order of Coleoptera²², (ii) the genome and transcriptome sequences are well annotated for *T. castaneum* species^{23,24} and, (iii) lack of genetic resources of *A. sputator* species in the databases.

Three *Agriotes* species, *A. obscurus*, *A. lineatus* and *A. sputator*, are known to exist in the Atlantic provinces of Nova Scotia, New Brunswick, Newfoundland and Prince Edward Island (P.E.I.)¹ of Canada. The morphology of these three species of *Agriotes* is remarkably similar and challenging to distinguish visually. Due to a lack of precise distinguishing characteristics among the *Agriotes* species, it is important to identify which species and populations are present in each area. In this study, *A. sputator* species was confirmed by aligning *mtCOI* gene from RNA-Seq data of all five groups with the reference sequence of the partial *mtCOI* gene from NCBI database. The result showed that all the five groups of samples used in this study were *A. sputator* with 98% identity, and it suggested that *A. sputator* species is predominant and widely distributed in Eastern Canada. Vernon and van Herk²⁵ also reported that *A. sputator* is the highest in the field of potatoes in Eastern Canada using pheromone-based trap approach. Previous studies reported using the *mtCOI* gene to differentiate the *Agriotes* species. Staudacher et al.²⁶ sequenced the *mtCOI* gene from 17 *Agriotes* species in central Europe and developed multiplex PCR assays to identify nine most common species.

After confirmation of the samples that are indeed the *A. sputator* species, the differential gene expression analysis was performed. In order to identify DEGs during the various development stages of *A. sputator*, the female libraries were utilized as a reference to fully illustrate the comparisons in this work, between sexes (male and female click beetles) and among three larva stages based on their sizes. A total of 34,709 significant DEGs ($\log_2$ fold change > 2, p_{adj} < 0.05) were identified across the developmental stages including 7,712 DEGs (5,413 were known genes and 2,299 were novel genes), 9,219 (6,420 were known genes and 2,799 were novel genes), 9,554 (6,548 were known genes and 3,006 were novel genes), and 8,224 (5,304 were known genes and 2,920 were novel genes) found in LE, LM, LL, and AM, respectively, when compared with the AF stage. To evaluate the consistency and variance of the different developmental stages, PCA analysis (using all gene transcripts) was performed. Results showed three clusters, one cluster for the AF stage, second cluster for the AM stage, and third cluster for all three larva stages (LE, LM and LL) (Fig. 2). This finding suggested that the expressed genes in the wireworm larva stages have a different gene expression pattern than the adult click beetles.

In this work, the significant differentially expressed genes in the development of stages were enriched in KEGG pathways, including metabolic pathways, carbohydrate metabolism, biosynthesis of amino acids, lysosome, ubiquinone and other terpenoid-quinone biosynthesis, and these pathways play important roles in different stages of insect development¹⁶. Studies have shown that pathways related to carbohydrate and sugar metabolism provide the energy for other metabolic pathways are crucial for maintaining proper growth and development in insects²⁷. Similarly, the KEGG analysis of *Nicrophorus orbicollis* and *Hypothenemus hampei* at different development stages showed that functional pathways are mainly related to carbohydrate metabolism, the immune system, and signal transduction^{28,29}. Amino acids, as the building blocks of protein, also provide energy for metabolic pathways in insects^{30,31}. Lysosomes maintain cell homeostasis by regulating intracellular degradation, circulation, and signal transduction³².

In the LE and AF stages, DEGs were significantly involved in the metabolic pathways, neuroactive ligand-receptor interaction pathway, terpenoid backbone biosynthesis, purine metabolism, biosynthesis of amino acids, carbon metabolism, steroid biosynthesis, and toll and Imd signalling pathway. All DEGs, including serine protease (TRINITY_DN8998_c1_g1_i17), trypsin-1 (TRINITY_DN3955_c0_g1_i31), chymotrypsin-1 (TRINITY_DN395_c0_g1_i16), and chymotrypsin-2 (TRINITY_DN222_c1_g2_i1), which are involved in the neuroactive ligand-receptor interaction pathway were found to be up-regulated in LE compared to AF stage. In general, insect development and homeostasis are regulated by the neuroactive ligand-receptor interaction pathway, which comprises a collection of receptors on the plasma membranes and transduces various signals, including biogenic amines, neuropeptides, and lipoproteins^{33,34}.

This research revealed interesting gene expression variation across the *A. sputator* life cycle by comparing the reproductive and nonreproductive stages. In holometabolous insects, developmental control, and the transition from larva to adult are crucial from both evolutionary and ecological standpoints and for the development of long-term pest management strategies³⁵. This study describes expression differences between larval and adult stages and identified genes that could be linked to insect development. Interestingly in the LM stage, a few DEGs are directly associated with glutathione metabolism and, metabolism of xenobiotics by cytochrome P450. P450s are well known for using hydroxylation/oxidation to metabolize toxic substrates, making them one of the most rapidly radiating groups of detoxifying enzymes^{36,37}. Insect growth, development, reproduction, biosynthetic pathways of endogenous compounds, metabolism of insecticides and plant toxic chemicals, detoxification process and defense response against exogenous substance are regulated by P450 proteins^{38,39}. Some putative cytochrome P450-related genes were identified in the LE stage, which are involved in the oxidation-reduction process. In the LE stage, glutathione S-transferase 1-1 (TRINITY_DN10033_c0_g1_i1), and aminopeptidase N (TRINITY_DN4417_c0_g1_i2) genes were significantly up-regulated as compared to adult female beetles. In addition, genes acyl-CoA Delta (11) desaturase (TRINITY_DN4869_c0_g1_i5), fatty acid synthase (TRINITY_DN7102_c0_g1_i1), fatty acid synthase-like (TRINITY_DN1182_c0_g1_i2) related to the biosynthesis of fatty acids and fatty acid metabolism were up-regulated as well as genes fatty acyl-CoA reductase (TRINITY_DN3209_c0_g1_i4) and elongation of very long chain fatty acids protein (TRINITY_DN7358_c0_g1_i1) related to fatty acid elongation. In general, fatty acids are crucial compounds for insect development. They have a role in energy storage, cell and cuticle structure and production of pheromones⁴⁰. The major fatty acid groups in insects are associated with triacylglycerol as a source of metabolic energy, phospholipids, and sterol ester components for cellular and subcellular membranes⁴⁰. Phospholipid synthesis requires a variety of cofactors and enzymes, which are important components of life, and some phospholipids are essential nutrients⁴¹. In this study, the DEGs involved glycerophospholipid metabolism, including pancreatic triacylglycerol lipase (TRINITY_DN3441_c0_g1_i2), and lipase C (TRINITY_DN14056_c0_g1_i1) genes that were found to be significantly up-regulated in the LM stage.

In both LM and LL stages, DEGs are directly associated with the Forkhead box protein O (FOXO) signaling pathway, which is involved in developmental and longevity processes. The genes that are associated with the FOXO signaling pathway such as G2/mitotic-specific cyclin-B3 (TRINITY_DN3086_c0_g1_i7), catalase-like (TRINITY_DN589_c0_g1_i11), and serine/threonine-protein kinase polo (TRINITY_DN32289_c0_g1_i5) were found to be down-regulated as compared to the AF stage. Various studies revealed that the FOXO acts as the terminal transcription factor of the insulin signaling pathway and regulates a variety of physiological processes in many organisms, including lifespan in insects⁴².

Since wireworms and adult click beetles can overwinter during their life cycle, they must use one of the most fascinating strategies to survive the winter that is known as diapause. Diapause is a state of dormancy where metabolic and developmental arrested which are induced by adverse environmental conditions. This arrest occurs in a species-specific life stage, and its expression is regulated by various environmental signals that precede and reliably predict the arrival of unfavorable conditions⁴³. Many insect species overwinter as larvae/nymphs, during which the molting process is typically arrested, and some associated signaling pathways, such as hormone and energy metabolism, were suppressed^{44,45}. Few invertebrates, including *Drosophila melanogaster*, *Culex pipiens*, and *Caenorhabditis elegans*, depend on the insulin signalling system to regulate diapause^{46–48}. FOXO is the main transcriptional effector of the insulin signalling pathway and is typically suppressed in the presence of insulin⁴⁹. The phosphoinositide 3-kinase (PI3K)-serine/threonine kinase (Akt) pathway is triggered by elevated insulin levels, leading to the phosphorylation of FOXO and the subsequent inactivation of FOXO. In this study, the gene expression analysis result suggests that diapause enables both larval (LM and LL) stages of the wireworm to survive under adverse environmental conditions by down-regulation of the FOXO signaling pathway and up-regulation of laminin subunit alpha 4 (TRINITY_DN7714_c0_g1_i8) gene which is directly associated with PI3K/Akt pathway. It is thus fundamentally important to understand the appropriate diapause conditions of wireworms to optimize biocontrol programs against insect pests. Research has shown that protein phosphorylation and dephosphorylation under low temperatures and freeze-thaw conditions are significant regulatory mechanisms connected to various metabolic functions. The activity of numerous transcription factors can be controlled by protein kinases and phosphatases, and they can also take part in the differential expression of genes that are involved in cold and freeze resistance⁵⁰. Furthermore, Wnt signaling pathways are essential in cell growth and tissue patterning. In the last few years, exciting advances have been made regarding the molecular and cellular mechanisms involved during sequential segmentation and axial elongation in a variety of arthropods and the Wnt pathway has emerged as an essential factor in these processes^{51–53}. In this study, a few genes were found that were involved in Wnt signaling pathway such as E3 ubiquitin-protein ligase SIAH (TRINITY_DN7844_c0_g1_i5), and E3 ubiquitin-protein ligase sina-like (TRINITY_DN19338_c0_g1_i3).

In addition to comparing the reproductive and nonreproductive stages, the gene expression between the AM and AF also varied. Several key genes were involved in the process of spermatogenesis⁵⁴, including sperm-associated antigen 6-like TRINITY_DN4675_c0_g1_i1), tubulin beta-4 chain-like, outer dense fiber protein 2 (Odf2) (TRINITY_DN1014_c0_g1_i1), spermatogenesis-associated protein 17-like (TRINITY_DN7384_c0_g1_i1), deleted in azoospermia-like isoform X3 (DAZ), kelch-like protein 10 (TRINITY_DN6566_c0_g1_i9), armadillo repeat-containing protein 4 (TRINITY_DN3950_c0_g1_i2), and T-complex-associated testis-expressed protein 1 (TRINITY_DN14187_c0_g1_i2) and these genes were found to be up-regulated in the AM stage. The Odf2 gene is involved in the formation and maintenance of the sperm tail structure. It helps assemble the sperm tail axoneme, which is essential for sperm motility. Additionally, the DAZ gene and its family play an important role in meiosis (reductional maturation divisions) in the spermatogenesis of insect males⁵⁵.

In the process of oogenesis, the insect mother deposits lipids, proteins, carbohydrates which play a crucial role in the development of the embryo's posterior region. Particularly in oviparous species the insect egg must

contain all nutrients required for embryonic development⁵⁶. Interestingly, several genes related to vitellogenesis such as vitellogenin (TRINITY_DN80874_c0_g1_i3), vitellogenin 1 (TRINITY_DN220306_c0_g1_i1), vitellogenic carboxypeptidase (TRINITY_DN182281_c0_g1_i1), and cathepsin B (TRINITY_DN3250_c2_g1_i1) were significantly up-regulated in the AF stage while these genes were found to be down-regulated in the AM stage. Vitellogenin is a crucial protein involved in the reproduction of insects, particularly in females. It is synthesized in the fat body and is essential to produce eggs in oviparous species serving as a precursor to the egg yolk proteins⁵⁷. In addition, the protein claret segregational-like (TRINITY_DN20045_c0_g1_i1) which is encoded at the claret locus in insects is needed for normal chromosome segregation in meiosis in females and in early mitotic divisions of the embryo⁵⁸ and this protein was found to be up-regulated in the AF stage. In insects, sexual reproduction is a vital physiological process essential to population maintenance. Consequently, it is crucial to identify the genes involved in reproduction because doing so would aid in pest management. Evaluating the molecular mechanisms for higher-order insect species will also be helpful. Similarly, transcription initiation factor IIA subunit 1-like (TRINITY_DN10658_c0_g1_i2), and T-box transcription factor TBX20-like isoform X2 (TRINITY_DN4903_c0_g1_i4) genes were significantly up-regulated in males compared to AF, which were involved in the basal transcription factors pathway. In general, the transcription factors cooperate with RNA polymerase II to bind and open promoter DNA to initiate RNA synthesis and stimulate the escape of RNA Pol II from the promoter. RNA polymerase II assembles with the general transcription factors (GTFs) at promoter DNA to form the pre-initiation complex for initiation.

Using high-throughput screening, several candidate genes could be tested from this present study to develop efficient RNAi-mediated pest control methods. For instance, CYP450s-related genes could be selected for an RNAi target as they play an essential physiological role in the growth, development, and reproduction of insects^{59,60} and are the most widely studied for RNAi^{11,61,62}. For example, carbonyl reductase [NADPH] 1, glutathione S-transferase 1–1, glutathione S-transferase 1 and UDP-glucuronosyltransferase 2B14-like genes. Furthermore, insulin and Wnt signalling pathways-related genes could be useful for RNAi research on *A. sputator* which are associated with the diapauses process and involved in essential cell growth and tissue patterning, respectively. For instance, *FOXO*, insulin receptor substrate 1, catalase-like, cyclin-dependent kinase inhibitor 1 A (cdkn1a), and RAC serine/threonine-protein kinase-related genes, which are associated with insulin signalling pathway while E3 ubiquitin-protein ligase SIAH, and E3 ubiquitin-protein ligase sina-like genes that are involved in Wnt signalling pathway, could be used for RNAi studies.

In this study, several differential expressed genes have been identified which play a crucial role in insect female reproduction such as vitellogenin, vitellogenin 1, and vitellogenic carboxypeptidase and these could be selected as the feeding of dsRNA for RNAi approach. Many previous studies also reported that the knockdown of vitellogenin-related genes using RNAi method hampered the ovarian development in *Chrysopa pallens*⁶³, inhibited yolk deposition in *Colaphellus bowringii*⁶⁴ and halted the reproduction of *Rhynchophorus ferrugineus*⁶⁵.

Conclusions

We have generated a transcriptome assembly of the *A. sputator* species in five different click beetle developmental stages namely the early, mid and late-stages of larva and adult female and male click beetle. A substantial number of genes related to development pathways, general function, and metabolic processes are found in these transcriptome assemblies. Genes that exhibit differential expression at various stages of their development were also found. This is the first report of transcriptome sequencing in the coleopteran insect pest wireworm *A. sputator*. This information will be essential for the *de novo* assembly of its genome sequence. The transcriptomic data generated from this study also offers studying wireworm biology and potential molecular targets for RNA interference (RNAi)-based wireworm control. Finally, the work may contribute to understanding the molecular mechanisms behind wireworm insect development and reproduction.

Materials and methods

Insect sample collections

Wireworms were collected from the potato, carrot, and cabbage fields in Charlottetown, Prince Edward Island (P.E.I.), Canada. They were reared in the laboratory on moist soil (~5 cm deep) with pieces of potatoes or carrots in a well-aerated container at 22 ± 2 °C (room temperature, RT), 60% RH in dark cabinets until further experiments. The adult male and female click beetle samples (*A. sputator*) were collected from Agriculture and Agri-Food Canada Charlottetown Research and Development Centre, P.E.I., and samples were frozen in liquid nitrogen and stored at -80 °C until RNA extraction. Morphologically, wireworm larvae range from white to shiny golden brown with advancing developmental stages and exhibit a size range of 2 to 18 mm long. Wireworms have a cylindrical and smooth-segmented body. The anterior end (dark brown) comprises the head, three pairs of legs, and biting mouthparts⁶⁶. The last abdominal segment of the body is narrowing, with two small dark spots on the upper surface which is a distinguishing feature of *Agriotes* wireworms⁶⁷. Adult beetles of *A. sputator* have a dark brown to the chestnut brown elongated body, variable in size (6–8.5 mm long and 2 mm wide). These morphological features were considered to select the species of interest and identify them in different developmental stages for the current study.

RNA isolation and Illumina sequencing

Since the instar of wireworm is unknown, three different larval size groups based on their body length and one each adult female and male click beetles were selected for this study. The five groups included the early-stage larvae (LE, size range from 9 to 10 mm); the mid-stage larvae (LM, size range from 12 to 14 mm); the late-stage larvae (LL, size range 16–18 mm); adult female click beetle (AF) and adult male click beetle (AM). In total, 20 samples that included five developmental stages and four biological replicates (early-stage larvae: LE1, LE2,

LE3, LE4; mid-stage larvae: LM1, LM2, LM3, LM4; late-stage larvae: LL1, LL2, LL3, LL4, adult female click beetle: AF1, AF2, AF3, AF4 and adult male click beetle: AM1, AM2, AM3, AM4) were selected for RNA extraction and cDNA sequencing. Five individual larvae from each of the three stages were pooled together to produce one replicate, while one adult female and one adult male click beetle was used as one replicate. Total RNA was isolated from each developmental stage using RNAeasy Mini Kit (Qiagen, CA, USA), following the manufacturer's protocol, and stored at -80°C until use. DNase digestion was performed using an RNase-free DNase (Qiagen, CA, USA). The RNA purity (OD 260/280) and quantity were assessed using NanoDrop ND-1000 UV-Vis Spectrophotometer, and Qubit RNA assay kit, broad range (BR) in the Qubit 2.0 Fluorometer (Life Technologies, CA, USA), respectively. In addition, the contamination and degradation of RNA were checked using 1% agarose gels. The quality and integrity of RNA [RNA integrity number (RIN)] was also confirmed using a 2100 Bioanalyzer (Agilent Technologies, USA) at Genome Quebec (Montreal, Quebec, Canada). cDNA libraries were prepared using the NEBNext Ultra II RNA library preparation kit. After quality assessment, RNA sequencing was performed using the Illumina NovaSeq6000 platform at Genome Quebec (Montreal, Quebec, Canada). Paired-end 100 bp reads were generated for each sample.

Quality check and trimming

The quality of raw sequencing data was checked using FastQC v.0.11.2⁶⁸ (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>). The reads were evaluated for average read length, adapter content, per-sequence GC content, and sequence quality scores. The adapters and low-quality reads near the 3' and 5' ends were removed using Trimmomatic v.0.3.2⁶⁹ (<https://github.com/usadellab/Trimmomatic>). Reads were scanned with a sliding window of 4 bases wide; leading or trailing bases with an average Phred quality score of less than 35 were eliminated⁶⁹.

de novo transcriptome assembly

As there is no reference genome for *A. sputator*, *de novo* transcriptome assembly was performed using the Trinity v2.0.6⁷⁰ (<https://github.com/trinityrnaseq/trinityrnaseq>), which generated a set of components containing a variable number of transcripts (isoforms). Trinity comprises of three modules: Inchworm, Chrysalis, and Butterfly. First, Inchworm builds a K-mer dictionary from the reads, leading to the construction of the contigs. Next, Chrysalis connects all the overlapped contigs into components using a de Bruijn graph algorithm. In the final step, Butterfly simplifies all the generated graphs to report full-length transcripts and their alternatively spliced form. The analysis was performed using default parameters. To statistically assess the quality of the assembled transcriptomes, we used assembled transcripts as reference sequences. All clean reads of each sample from Illumina sequencing were mapped back to the assembled transcriptome using Bowtie2 v0.12.6⁷¹ software (<https://github.com/BenLangmead/bowtie2>).

Transcriptome functional annotation

Transcripts were annotated using NCBI-BLAST (National Center for Biotechnology Information – Basic local alignment and search tool) nonredundant protein (NR) database using the blastx algorithm with an e-value cut-off set to $1\text{E}-22$ using BLAST + 2.11.0⁷² (<https://ftp.ncbi.nlm.nih.gov/blast/executables/blast+/LATEST/>). Additionally, Gene Ontology (GO) assignment programs were utilized for the functional categorization of annotated genes using DAVID (Database for Annotation, Visualization and Integrated Discovery)⁷³ and functional pathways from the KEGG (Kyoto Encyclopedia of Genes and Genomes, <https://www.genome.jp/kegg/>)⁷⁴. To confirm *A. sputator* species in a given area, the *mtCOI* gene was retrieved from RNA-Seq data of all five stages as *COI* gene is a reliable mtDNA marker for differentiating between all species^{76,77} and the sequences were aligned to a reference sequence of *mtCOI* gene (NCBI Accession number: KM445546.1) using ClustalW (Bioedit)⁷⁸. A phylogenetic tree was constructed based on partial *mtCOI* sequences of all five groups with the reported 12 *Agriotes* species using the maximum likelihood approach using MEGA11⁷⁹ (Supplementary Fig. 2).

Gene expression estimation

To compare gene expression profiles across the three wireworm larva stages (LE, LM, and LL) and the two adults stages (AF and AM) of *A. sputator*, the gene expression level was measured with Salmon⁸⁰ (<https://github.com/COMBINE-lab/salmon>). Salmon quantifies the expression of transcripts using quasi-mapping with a two-phase inference procedure algorithm to provide accurate expression estimates. The relative abundance of each transcript for each sample was calculated into Transcripts Per Million (TPM).

Differentially expressed genes (DEGs) and pathway analysis

Prior to differential gene expression analysis, the read counts were filtered out for quality control based on the following criteria: (i) transcripts with zero counts in all samples and, (ii) transcripts that were not present in at least 50% of the samples^{81,82} while comparing two groups using the edgeR⁸³ program package (<https://bioconductor.org/packages/release/bioc/html/edgeR.html>). DESeq2 v3.16⁸⁴ (<https://bioconductor.org/packages/release/bioc/html/DESeq2.html>) was used which is based on Negative-binomial distribution to identify the DEGs across the developmental stages. The p-values were adjusted (padj) using Benjamini and Hochberg correction, which controls the false discovery rate (FDR). The DEGs with a p-value adjusted (padj) < 0.05 and $\log_2$ foldchange ≥ 2 were considered significant. The adult female group was considered as a reference and DEGs that are up- or down-regulated were classified based on the $\log_2$ fold change direction. The up-regulated and down-regulated genes were visualized using a volcano plot constructed using the R-package EnhancedVolcano v1.8.0⁸⁵. The principal component analysis (PCA) plots were created using DESeq2 v3.16. GO and KEGG enrichment pathways of DEGs were performed by the DAVID database⁷³ (<https://david.ncicrf.gov/home.jsp>). Pathways

with $\text{padj} \leq 0.05$ were considered significant. The bar charts were constructed using online SRplot tool (<https://www.bioinformatics.com.cn/en>)⁸⁶.

Data availability

Raw sequencing reads were deposited into NCBI- Sequence Read Archive (SRA) database under accession number PRJNA977396 but access to these data is restricted. Once this research is published, the data will be made available to the public.

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Author contributions

G.W.P. and J.J. designed the project. J.J. performed the experiments and analyzed the data. J.J. and G.W.P. wrote the manuscript. All authors read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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